

EDUCATION

Carnegie Mellon University <i>M.S. in Software Engineering</i> GPA: 4.0 / 4.0 Selected Coursework: Machine Learning, Large Language Model Systems, Data Science for Software Engineering, Distributed Systems	Expected May 2027
University of Wisconsin–Madison <i>B.S. in Computer Science & B.S. in Computer Engineering</i> GPA: 3.72 / 4.0 Selected Coursework: Large Language Models in Practice, Software Engineering, Algorithms	May 2024

WORK EXPERIENCE

NewsBreak <i>AI Engineer Intern</i>	May – August 2026 <i>Mountain View, CA</i>
- Designed and shipped a personalized LLM alert pipeline as a Kubernetes CronJob (embed → LLM-classify and score → per-topic percentile select → LLM same-story dedupe → push; ~1,200 LLM calls/day), logging why each item fired or did not. - Evaluated it over 11 days of production shadow on 4,100+ logged decisions: hand-graded alerts showed filler and real events both scored 89–90, so added a categorical <i>happened</i> / <i>speculative</i> / <i>no_event</i> gate, checked it on a 127-alert holdout, and ran prompt ablations; rejected a rule that caught 91% of filler but blocked the day’s two highest-reach real alerts; shipped live with filler 34% → 4%. - Shipped RAG and agentic search on one SSE endpoint: turn 1 is deterministic RAG over hybrid retrieval (substring + text + vector, fused by RRF); follow-ups run a bounded tool-calling agent (≤3 hops, 5 tools, 45 s) with server-resolved citations the model cannot author. - Built an eval harness on real logged queries against the committed catalog; deterministic entity resolution took P@1 from 0% to 100% (28/28 recall) at 0.4 ms, after measurements ruled out embeddings. - Diagnosed a 65% failure rate in another team’s LLM pipeline via Langfuse traces as an unbounded reasoning loop (p95 1,197 tokens vs. 16,384 runaways) and sized a 2,048-token cap against observed successes. - Built the FastAPI / MongoDB backend from the first commit to 89 endpoints on AWS EKS, with ~1,450 fully-offline tests behind a ruff / pyright / pytest CI gate; wrote 34 contract-first API specs. - Sole SwiftUI engineer for SportsBreak (~6,000 installs) and primary backend author; 344 PRs, 340 merged.	
Haddee Education <i>Engineering Lead</i>	June – September 2025 <i>San Diego, CA</i>
- Designed database schemas and REST APIs for LangChain LLM features: resume parsing, improvement suggestions, and AI job matching. - Led an 8-intern team through Agile sprints to ship a React, Next.js, FastAPI, and Supabase web app on GCP App Engine; led PR reviews and mentored interns on Git workflows and testing.	
ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch <i>Full Stack Developer</i>	September – December 2023 <i>Beijing, China</i>
Sole engineer on a VC-backed startup’s LLM mock-interview web app (React, Next.js, Supabase); worked with recruiters so AI scoring categories matched real hiring, and iterated scoring with clients until three 200–500-person companies signed contracts.	

SKILLS

AI & ML	RAG, hybrid retrieval (RRF), LLM tool calling / agents, LangChain, Langfuse, PyTorch
Languages	Python, SQL, TypeScript, Go, Java, Swift
Testing & Practice	pytest, Unit Testing, TDD, ruff, pyright, Git, CI/CD, Jira, Agile/Scrum
Backend & Data	FastAPI, Pydantic, REST APIs, Server-Sent Events (SSE), PostgreSQL, MongoDB
Cloud	AWS, GCP Vertex AI, Kubernetes (EKS), Docker, Helm, GitHub Actions

PROJECTS

- my-av:** Built a PyTorch CNN-GRU that predicts steering angle and speed from 1 s video; trained on 33 hours of Comma2k19 on local CUDA and GCP Vertex AI, with streaming inference.
- codingplans:** Building a site that rates LLM coding plans; bought Z.AI, MiniMax, and Kimi plans and tested KV caching, long context, and needle-in-a-haystack to expose nerfed caching and heavy quantization.